The word response is not constant: variability, task drivers and memory

Research proposal (draft)

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Summary

Studies of free recall compare words that are later recalled with words that are forgotten, treating the response to the word itself as a constant that subtracts out. In 362 neurosurgical patients performing delayed free recall, the word response is not constant. It differs between electrodes of the same region in a way that replicates across sessions, its alpha and high-gamma components vary independently across electrodes, and from trial to trial it scales with where the word falls in the list. Serial position changes the size of the high-gamma response but not its timing. Because serial position and list number also predict recall, they account for much of the subsequent memory effect, as recently shown for the recall contrast itself (DeHaan, 2026). We propose to characterize where the variability of the word response sits, what drives it from trial to trial, and which part of it carries information about later memory.

Background and gap

Serial position and the subsequent memory effect. High-frequency activity during encoding falls across serial positions regardless of later recall (iEEG, encoding; Serruya et al., 2012); posterior gamma predicts recall mainly for early items (scalp EEG, encoding; Sederberg et al., 2006). Scalp EEG classifiers match recalled and forgotten words across serial positions (encoding and retrieval; Li et al., 2024), and in fMRI no signal predicts later memory once serial position and item are accounted for (encoding; Halpern et al., 2023). In 377 iEEG subjects, adjusting for serial position, item memorability and list number attenuates the subsequent memory effect by about 40% across regions and frequencies, mostly through serial position (encoding; DeHaan, 2026). These studies model the recalled vs. forgotten contrast; none characterizes how the same factors change the response to the word itself, its timing, or the blank interval before it.

Blank interval. Longer gaps before an item improve its recall (behaviour; Brown et al., 2006). A response after a longer wait can be larger because attention is directed to the expected moment (scalp EEG, cued timing; Correa et al., 2006) or because it has recovered more from the previous stimulus; expected stimuli can also give smaller responses (MEG, auditory; Todorovic and de Lange, 2012).

Heterogeneity. High-frequency activity is suppressed in default-mode regions during tasks (iEEG, visual search vs. rest; Ossandón et al., 2011), and sites within a region increase or decrease

during arithmetic (iEEG, arithmetic vs. rest; Pinheiro-Chagas et al., 2024). Heterogeneity alone is expected in visual cortex; what is not known is whether the electrode-level pattern of the word response is stable within a subject, whether its spectral components go together, and how large it is relative to differences between subjects.

Data

FR1, catFR1 and pyFR (362 subjects, 1009 sessions). Each word (word on, 0–600 ms) is compared with the blank screen before it (word off, −700 to −100 ms; the first word of a list follows the countdown and is excluded). Bands: alpha (8–13 Hz, multitaper) and high gamma (70–150 Hz, Hilbert envelopes). Trial-level analyses drop robust outlier trials per electrode (§Open issues).

Aim 1. Where and when does the word response vary?

Between subjects and electrodes. Subject distributions are wide in every region (Fig. 1). Splitting electrode-level Cohen's d into between-subject and within-subject (between-electrode) parts, subjects account for only 12–37% of the variance in high gamma and 25–36% in alpha; most of it lies between electrodes of the same region in the same subject. Regions mix directions (Fig. 2): 40% of occipital electrodes increase in high gamma and 3% decrease; in frontal cortex 5% increase and 4% decrease; in alpha, 21% of occipital electrodes decrease and 11% increase.

The electrode pattern is stable, and its two bands are independent. Within a subject, the pattern of responses across a lobe's electrodes replicates between session halves (Fig. 3, left; median $r = 0.90$ occipital, 0.59 parietal, 0.55 frontal high gamma; 0.20–0.80 for alpha), so the heterogeneity is not measurement noise. Across the same electrodes, alpha and high-gamma responses are essentially uncorrelated (right; median r between −0.10 and 0.19): the alpha decrease and the high-gamma increase are not two readouts of one activation but separate signals with separate spatial patterns.

Timing and network. High-gamma latency increases from lateral occipital cortex (∼120 ms) to orbitofrontal cortex (∼320 ms; Fig. 4), with adjacent regions largely overlapping. Alpha synchronization hubs move from occipital to parietal to temporal regions over the first second (Fig. 5). Power changes are reliable within subjects across sessions, synchrony changes are not, and across subjects only the occipital connectivity profile replicates (Fig. 6).

Conclusion. The word response is a stable, subject-specific pattern across electrodes rather than a regional average plus noise, with alpha and high gamma dissociated electrode by electrode. Its timing is ordered across regions and its network moves forward, but only the occipital network is reproducible across subjects.

Plan. (1) Variance components (subject, electrode, session, trial) per region and time window. (2) Map increasing and decreasing electrodes in MNI space within regions (frontal eye fields and lateral prefrontal vs. other frontal cortex; ventral vs. dorsal occipital) and by recording type. (3) Test the alpha–high-gamma dissociation per electrode over time (does alpha lag high gamma where both respond?). (4) Electrode-level latency: is latency variability within a region larger than between regions?

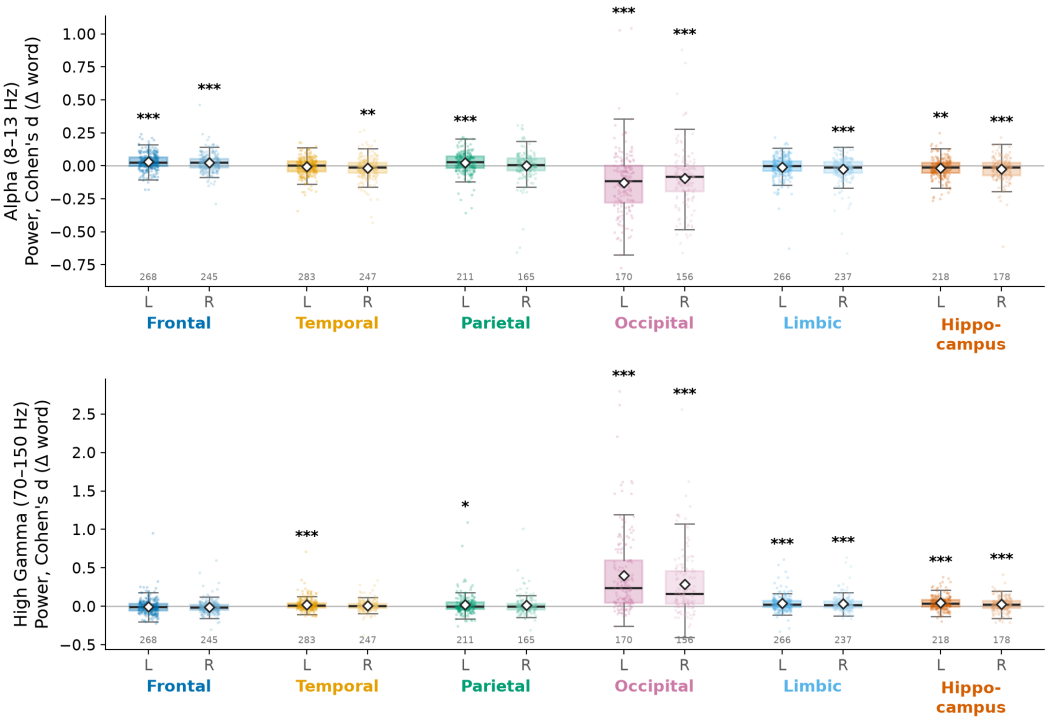


Figure 1: Word-onset power change per region, distributions across subjects.

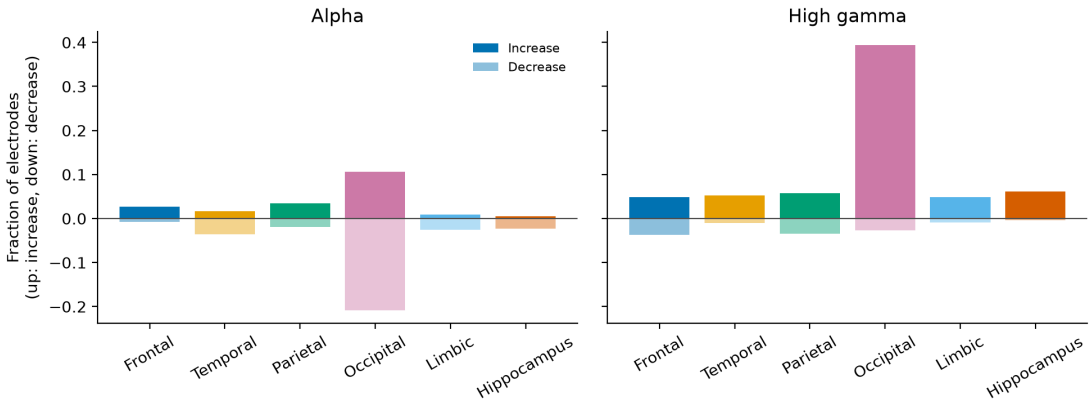


Figure 2: Fraction of electrodes whose power significantly increases (up) or decreases (down) at word onset (per session: Welch t on log power, word on vs. word off, BH-FDR across the session's electrodes).

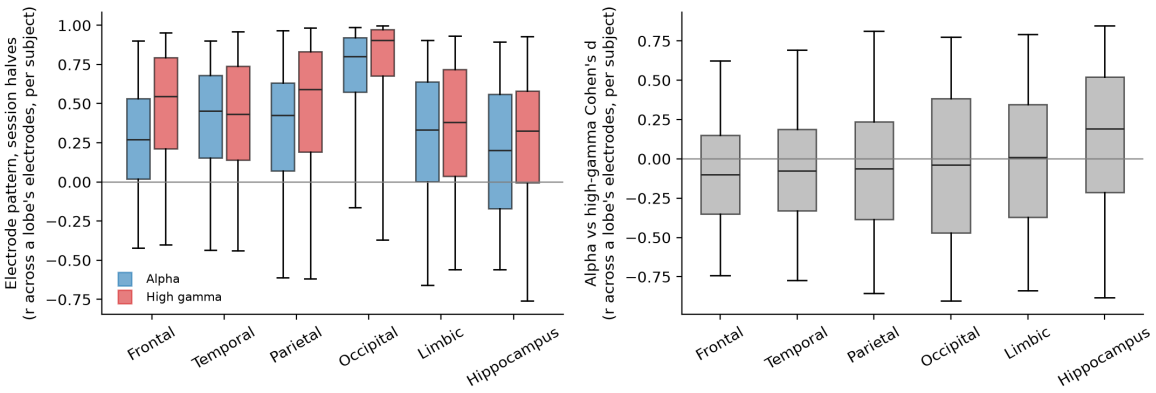


Figure 3: (Left) Stability of the electrode pattern: per subject and lobe, Pearson r across the lobe's electrodes between Cohen's d in two halves of the subject's sessions (subjects with $\geq$2 sessions and $\geq$5 electrodes in the lobe). (Right) Per subject and lobe, Pearson r across electrodes between alpha and high-gamma Cohen's d. Boxes across subjects; whiskers: central 95%.

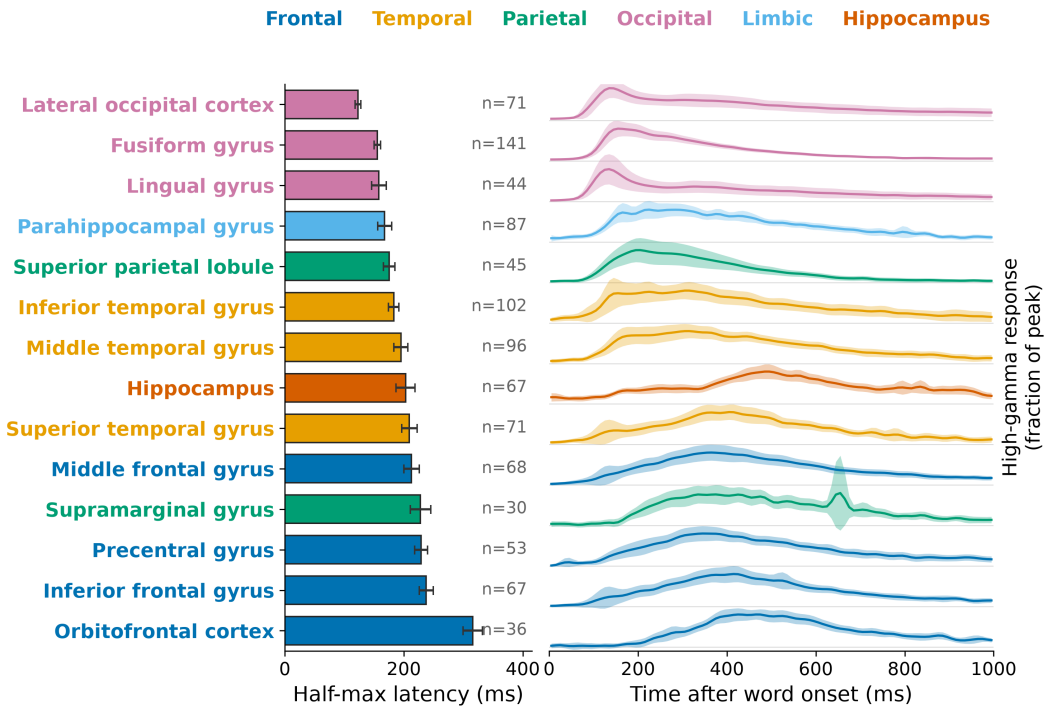


Figure 4: High-gamma half-max latency per anatomical region and the group response over time (from the word-onset paper). The spike at ~650 ms in supramarginal gyrus is one artifact trial in one subject (§Open issues).

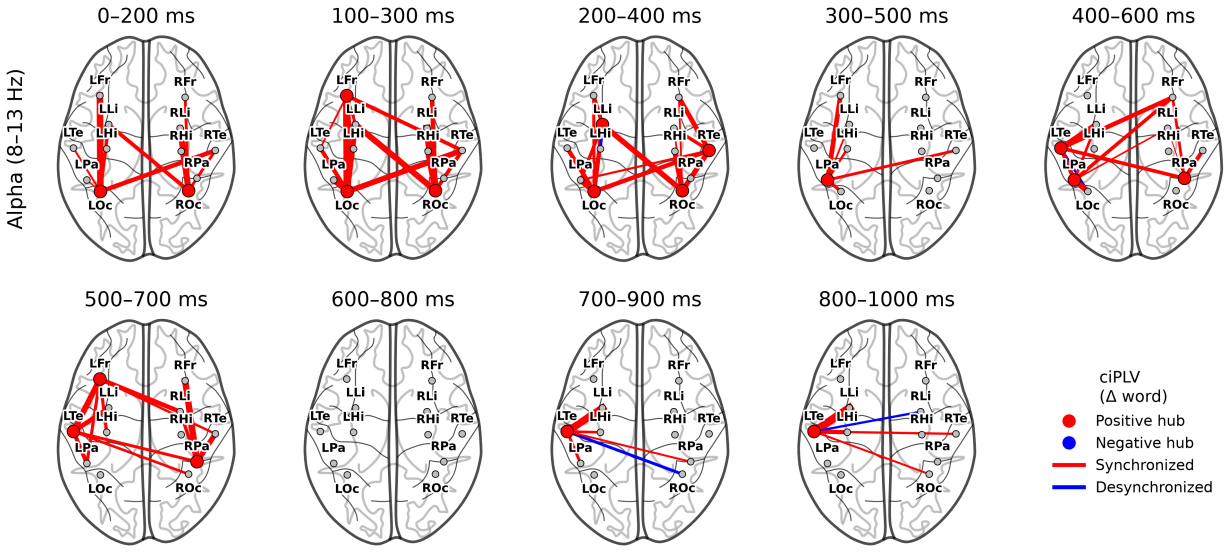


Figure 5: Time-resolved alpha ciPLV network in 200 ms windows (from the word-onset paper): red (blue) nodes are regions whose mean change over their connections is above (below) zero.

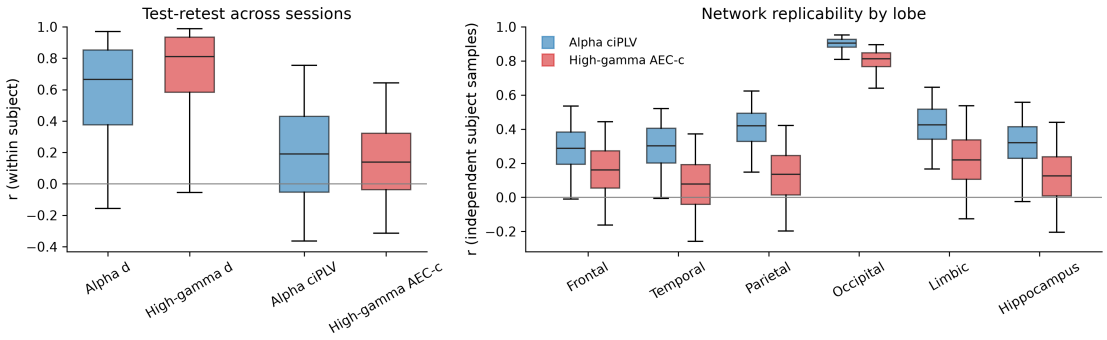


Figure 6: Session-to-session reliability within subjects, and replicability across subjects (from the word-onset paper).

Aim 2. What drives the word response from trial to trial?

Preliminary. Per subject and lobe, each trial's word-on power (log, z-scored within session) was regressed on serial position, list number and blank interval (Fig. 7). High gamma falls with serial position in every lobe (occipital −0.042 SD per SD of serial position, temporal −0.032, frontal −0.020; all $q < .05$, 254–339 subjects) and alpha rises (0.009–0.021), as reported for the recall contrast (Serruya et al., 2012; DeHaan, 2026). Alpha also rises across lists in most lobes. The serial-position effect is a change in gain, not timing (Fig. 8): the high-gamma peak is smaller for positions 8–12 than 2–4 in every lobe (paired $p < .001$; frontal 0.41 vs. 0.30, hippocampus 0.32 vs. 0.21), while half-max latency does not change in any lobe after correction over lobes (all $p > .18$ except limbic, $p = .02$). The blank interval had small effects (occipital high gamma +0.007, frontal alpha +0.006), consistent with the occipital-specific effect after longer waits (Fig. 9). Together the three variables explain 1–2% of trial-to-trial variance.

Conclusion. The word response is modulated by list structure: across a list it shrinks in high gamma and grows in alpha everywhere, without shifting in time, and it habituates across lists in alpha. Most trial-to-trial variation is not explained by these variables.

Plan. (1) Time-resolved and latency effects per region and electrode. (2) Separate expectation from recovery after the previous word: the previous word's response size and offset time as covariates, a hazard-rate vs. an adaptation model, and experiments with other timing (RepFR1, ltpFR2). (3) Whether serial-position effects shrink over lists and sessions. (4) Pre-stimulus alpha and high gamma as predictors of the unexplained variance.

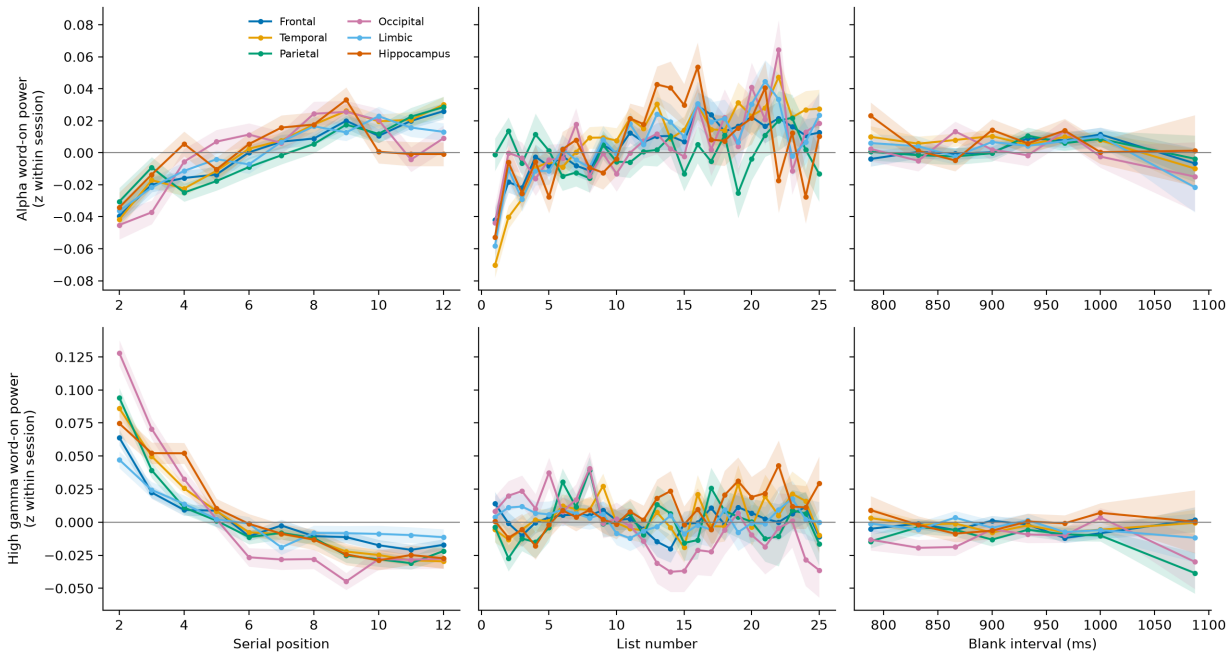


Figure 7: Word-on power (z within session) by serial position, list number and blank interval (quantile bins); mean ± SEM across subjects per lobe. Serial positions 2–12; scored lists only.

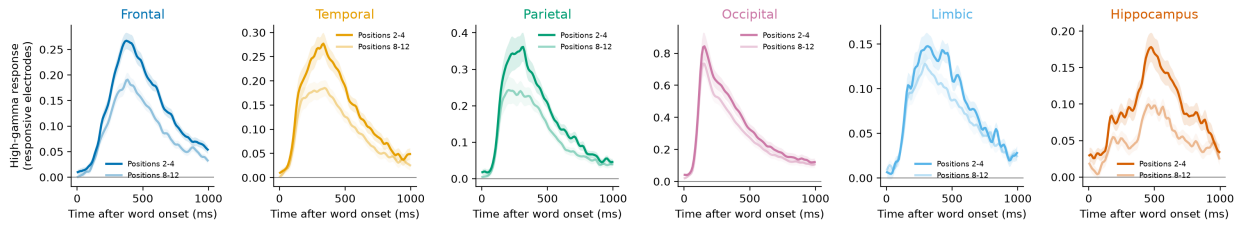


Figure 8: High-gamma response of responsive electrodes for serial positions 2–4 (dark) and 8–12 (light); mean ± SEM across subjects.

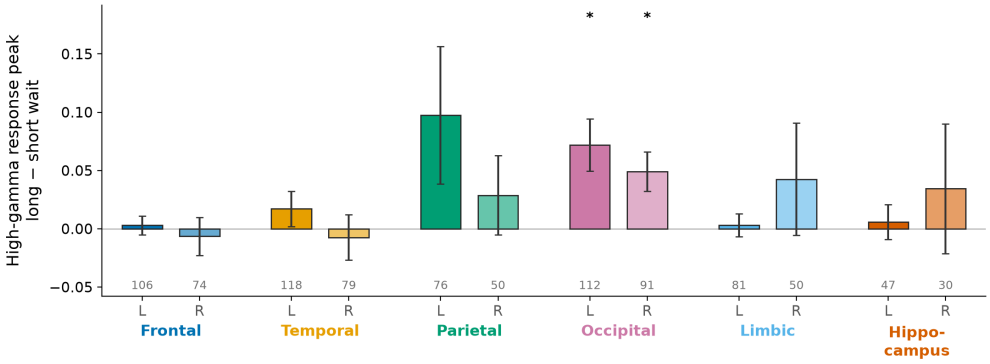


Figure 9: Occipital high-gamma response after long vs. short waits (from the word-onset paper).

Aim 3. Which part of the word response carries information about memory?

Preliminary. Serial position and list number predict recall (logistic $\beta = -0.19$ and -0.09 per SD, 361 subjects, both $p < 10^{-6}$); the blank interval does not ($p = .47$). Removing serial position (linear and positions 2–3), list number and blank interval from each trial's response (Fig. 10) reduced the high-gamma subsequent memory effect from $d = 0.030$ to 0.014 in frontal cortex, 0.020 to 0.001 in temporal cortex, 0.014 to 0.001 in the hippocampus, and 0.098 to 0.067 in occipital cortex (paired t, all $p < .01$), in line with DeHaan (2026). Time-resolved decoding of the response predicts recall weakly in every lobe (Fig. 11).

Conclusion. Within the word response, the part that predicts memory is largely the part that follows list structure; the effect that survives adjustment is concentrated in occipital cortex.

Plan. (1) Decode recall from the response with covariates removed and with recalled and forgotten words matched on serial position and list (as Li et al., 2024), time-resolved, to ask when the surviving signal appears. (2) Ask whether the surviving signal is a scaled version of the word response (the same electrodes and timing) or a different pattern, using the stable electrode patterns of Aim 1. (3) Compare electrodes suppressed and activated at word onset in frontal cortex. (4) Test whether the blank interval affects recall of the next or the current word and whether the word response mediates it.

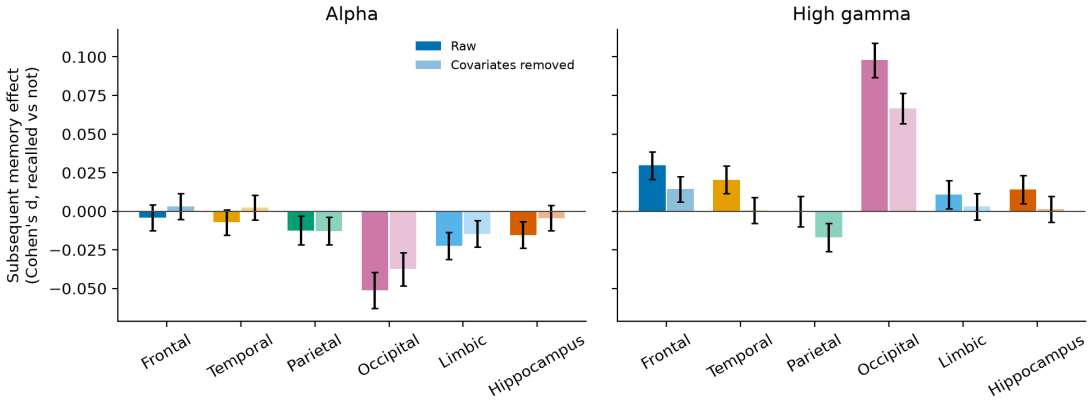


Figure 10: Subsequent memory effect (Cohen's d, recalled vs. not recalled, word on 0–600 ms) per lobe, raw (dark) and after regressing each trial's response on serial position, list number and blank interval within subject (light); mean ± SEM across subjects.

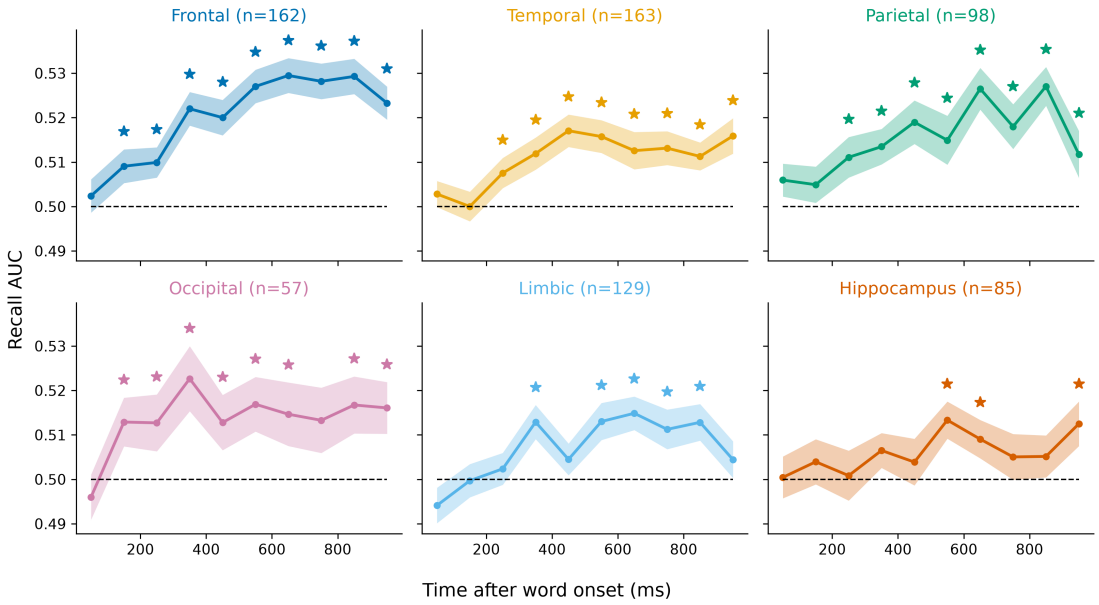


Figure 11: Recall prediction from the high-gamma response (encoding contrast; per-session AUC, mean ± SEM across subjects; from the word-onset paper).

Open issues and extensions

- **Artifact trials.** The main pipeline has no trial-level artifact rejection: one trial in one subject (a broadband transient on 82 of 148 electrodes at 655 ms) produces the supramarginal spike in Fig. 4. The preliminary results here drop robust outlier trials; the main analyses need the same.
- **Serial position 1** is excluded by the baseline design; primacy is measured from position 2.
- **Blank intervals** span only a few hundred ms; experiments with other timing would extend Aim 2.
- **Eye tracking** during word presentation (the Kucewicz datasets) would separate visual input (fixations, saccades) from the word response.

References

Brown, G. D. A., Morin, C., and Lewandowsky, S. (2006). Evidence for time-based models of free recall. *Psychonomic Bulletin & Review*, 13:717–723.

Correa, Á., Lupiáñez, J., Madrid, E., and Tudela, P. (2006). Temporal attention enhances early visual processing: A review and new evidence from event-related potentials. *Brain Research*, 1076:116–128.

DeHaan, R. D. (2026). *On the specificity, causality, and reliability of electrophysiological correlates of human memory: limitations and opportunities*. PhD thesis, University of Pennsylvania.

Halpern, D. J., Tubridy, S., Davachi, L., and Gureckis, T. M. (2023). Identifying causal subsequent memory effects. *Proceedings of the National Academy of Sciences*, 120(13).

Li, Y., Pazdera, J. K., and Kahana, M. J. (2024). Eeg decoders track memory dynamics. *Nature Communications*, 15(1).

Ossandón, T., Jerbi, K., Vidal, J. R., Bayle, D. J., Henaff, M.-A., Jung, J., Minotti, L., Bertrand, O., Kahane, P., and Lachaux, J.-P. (2011). Transient suppression of broadband gamma power in the default-mode network is correlated with task complexity and subject performance. *The Journal of Neuroscience*, 31(41):14521–14530.

Pinheiro-Chagas, P. et al. (2024). Spatiotemporal dynamics of successive activations across the human brain during simple arithmetic processing. *Journal of Neuroscience*, 44.

Sederberg, P. B., Gauthier, L. V., Terushkin, V., Miller, J. F., Barnathan, J. A., and Kahana, M. J. (2006). Oscillatory correlates of the primacy effect in episodic memory. *NeuroImage*, 32(3):1422–1431.

Serruya, M. D., Sederberg, P. B., and Kahana, M. J. (2012). Power shifts track serial position and modulate encoding in human episodic memory. *Cerebral Cortex*, 24(2):403–413.

Todorovic, A. and de Lange, F. P. (2012). Repetition suppression and expectation suppression are dissociable in time in early auditory evoked fields. *Journal of Neuroscience*, 32:13389–13395.